DGLink: automated knowledge graph construction from biomedical data repositories

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Introduction: Biomedical discovery increasingly requires integrating fragmented data sets with existing knowledge. Several data repositories maintained by funding agencies and private institutions aim to provide standardized access to data of multiple modalities and formats, including genomic sequencing, transcriptomics, proteomics or imaging. However, the discoverability, reusability and interoperability of this data crucially relies on semantic annotations describing its context and content. Providing such annotations is time consuming and often poses a significant bottleneck in the release of data. To overcome this challenge, we developed DGLink, an automated system to traverse studies and data sets in data repositories and construct semantic annotations corresponding to experimental conditions and readouts. Key to the approach is the flexible ingestion of diverse tabular data coupled to named entity recognition and normalization against biomedical ontologies containing genes, proteins and other experimental factors. In addition to annotating tabular data, DGLink supports extracting metadata from files following community standard formats such as Variant Call Format (VCF) and Digital Imaging and Communications in Medicine (DICOM), and is extensible to extracting annotations from further standard formats. These annotations are grounded in biomedical ontologies and are assembled into a knowledge graph (KG) that serves as a semantic interoperability layer across data sets and studies while also connecting data to external knowledge. DGLink also implements a semantic search functionality and project similarity metrics on the knowledge graph. We demonstrate DGLink on the Neurofibromatosis Data Portal [1] to automatically construct a knowledge graph from 310 disease studies yielding more than 20,000 nodes and 45,000 edges. Crucially, despite no standard format or schema, 96% of tabular datasets on the platform were successfully processed. This approach is generalizable to other biomedical data repositories facing similar semantic integration challenges. DGLink is available as an open-source Python package under the BSD 2-clause license at https://github.com/gyorilab/dglink.

Methods: DGLink's architecture (Figure 1) consists of three main components: data ingestion, entity extraction and normalization, and knowledge graph assembly. DGLink's key novelty is the ability to automatically extract and semantically normalize experimental context directly from heterogeneous, unstandardized data files at the scale of a data portal, in addition to ingesting any manually curated metadata available. This is achieved by crawling experimental files associated with each project and filtering them for user-specified file types. In the case of tabular data, these unstandardized files are then read with the Frictionless data framework [2] and validated using custom logic. DGLink then infers which columns of the table may contain biologically meaningful named entities, and links the entries of each selected column to standardized resources using the Gilda [3] named entity recognition and normalization system. DGLink also handles

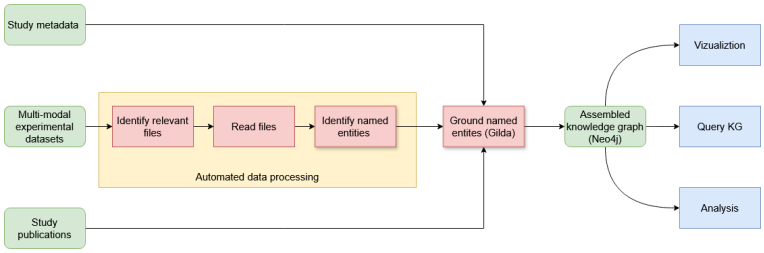

Figure 1: DGLink architecture

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standard formats including VCF and DICOM files, for genetic variant and imaging data, respectively, and will be extended to other common data formats. DGLink provides modules for extracting named entities from project metadata and associated publications. These modules parse the metadata and publications extracting both biologically meaningful entities (genes, small molecules, etc.) which are linked to standard ontologies, and non-biological named entities (project IDs, DOIs, etc.). The extracted metadata is represented as a set of nodes and relationships that are then assembled into a knowledge graph, and instantiated in a https://neo4j.com graph database.

Case study - NF Data Portal: To demonstrate DGLink's ability to create a semantic interoperability layer connecting multi-modal data in a data repository, we applied it to the Neurofibromatosis (NF) Data Portal [1] which houses experimental data from hundreds of NF disease studies. This portal contains project-level metadata and some annotation of the types of data in each project, however, the content of individual data files is not annotated, making the process of granular data search (e.g., which projects have tested EGFR-inhibitors), and connection of data to external knowledge (e.g., what publications describe the effect of EGFR inhibitors in NF diseases) limited. To complement the modules implementing the general knowledge graph construction approach described above, we implemented a client for the Synapse API to traverse projects and corresponding data files in the NF Data Portal, which uses Synapse as its back-end. This further allows applying DGLink to any collection of projects using Synapse. For the NF Data Portal DGLink traversed 310 projects, extracted project-level metadata, and processed $4,217$ tabular data files associated with these projects. Despite the lack of standard format or schema, DGLink was able to process 96% of these tabular files and identified $4,439$ individual columns containing biomedical entity terms. We evaluated three approaches to identifying relevant columns: a rule-based heuristic, a large language model-based approach and a combination of the two, and found our best performing method achieved an F1-Score of 0.82. Further, DGLink's entity normalization achieves 96% precision on text entries found in relevant columns. In addition to tabular data, we processed all available VCF and DICOM files. The knowledge graph constructed for the NF Data Portal consists of 28 node types and 28 edge types, the most common of which are shown in Table 1a, showing the range of biological information which was integrated. In total, the knowledge graph has $22,503$ nodes and $47,034$ edges. Table 1b shows the number of nodes and edges identified by each of DGLink's modules. The majority of nodes and edges were only identified in experimental data files, highlighting that DGLink's annotation process extends substantially on the otherwise available project metadata. Finally, we applied DGLink's project similarity search and semantic search features to the NF Data Portal knowledge graph. As an example to demonstrate semantic search, a researcher may ask, "What are the most common small molecules used across all Neurofibromatosis type 1 drug-screening studies?". Due to the lack of annotation in experimental files, using just the pre-existing NF Data Portal metadata would require manually checking hundreds of projects and thousands of files. Using DGLink's knowledge graph this question, can be formulated as a graph query and returns a list of $2,603$ small molecules and the number of projects studying them.

Entity type	Node count	Edge count
Gene/protein	12,317	21,671
Small molecule	5,785	9,342
Experimental factors	351	2,137
Disease	238	332

(a)

Data source	Node count	Edge count
Experimental data	18,328	32,481
Project metadata	3,151	11,173
Publications	200	229
Mixed	824	N/A

(b)

Table 1: 1a Most common node/edge types. 1b Node/edge sources.

Future work: This project has several promising directions of future work: 1) We will implement additional data repository API modules to allow application to a broader class of data repositories. 2) We will add additional handlers for standard data formats appearing frequently in data repositories. 3) We will integrate DGLink-created graphs with our INDRA literature-driven knowledge assembly system [4] as a starting point for AI-assisted data interpretation and hypothesis generation.

Declaration on generative AI: Claude was used for proof-reading during the writing process.

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